

arithmetic operations required for energy calculation

2. Suitable physical-layer communication interface for RF or power-line communication, if used

3. On-chip Flash memory and static random-access memory for storing application firmware and faster operation

4. LCD driver for LCD display

5. Interfaces like universal asynchronous receiver/transmitter, which can support infrared communication

6. High-resolution analogue-to-digital converters (ADCs) with programmable gain amplifier for voltage and current measurements

7. Input/output (I/O) ports for driving relays

8. Real-time counter for time keeping

The microcontroller senses the voltage and current through the signal conditioning circuit along with the ADC and programmable gain amplifier to calculate root mean square (RMS) voltage and current values, instantaneous energy consumed and total energy consumed over a period of time (one month or longer). It then sends this data to the utility meter through RF or power line communication and also displays it on the LCD. When a command to turn off the device is received, it drives suitable logic on its I/O ports to operate the relay.

The microcontroller gets its power supply from the power line through a 230V-3.3V converter. The converter can be suitably configured according to the operating voltage of the micro-

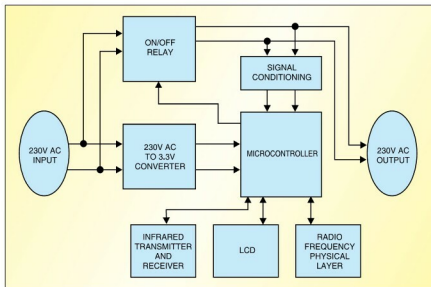


Fig. 2: Block diagram of HAN device

controller. The on-chip Flash firmware can be updated over the RF or PLC interface by the utility meter. The protocol and exact details of firmware update depend on specific implementations.

Signal conditioning. The signal conditioning unit consists of an analogue front-end for voltage and current measurement. The line voltage is measured by first down-sizing it with a resistor ladder, thereafter direct-current (DC) filtering and DC biasing.

Compared to voltage measurement, current measurement is less involved. First, the line current is down-sized using a current transformer and then passed through a small value of high-precision shunt resistor. The voltage drop across this shunt resistor gives a measure of the line current. As this voltage drop is very small, it is suitably amplified before being fed to the ADC.

The amplifier consists of programmable gain stages for amplification of only the alternating-current (AC) components, thus preventing the amplifier output from saturation.

Infrared interface. The infrared interface can be configured suitably according to the range and power consumption. It provides remote configuration support for the HAN

device, enabling the user to remotely turn on/off the home appliance connected to the HAN device. The protocol and exact details of operation can be flexibly chosen for particular implementations.

LCD panel. The LCD panel displays the instantaneous energy consumed, total energy consumed last/current month, date and time of the day, RMS voltage and RMS current. It inherits some of the utility meter display, thus acting as a low-accuracy but smart AC energy meter.

In a nutshell, the above architecture conceptualises a cost-effective and extremely versatile HAN device, which is replete with all the essential HAN device features along with the support for advanced features like firmware upgrade and full control of appliances over RF/PLC interface. It also doubles as a low-cost smart AC energy meter, providing round-the-clock energy consumption details of the home appliance.

Though the device is depicted here as a standalone intelligent power socket, it can also be implemented inside home appliances. ■

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